LINMA2171 — Homework 1

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1 Prony's method

(1.a)

Let $m \in \{0, \dots, M-1\}$. From eq.(2) and eq.(3), for all $k \in \{0, \dots, M\}$,

$$h_{k+m} = \sum_{j=1}^{M} c_j \, \mathrm{e}^{\omega_j (k+m)} = \sum_{j=1}^{M} c_j \, z_j^{k+m},$$

which is well defined since $0 \leq k+m \leq 2M-1$ and the $2M$ measurements $h_0, \dots, h_{2M-1}$ are available. Substituting this expression into eq.(5) and interchanging the two sums,

$$\sum_{k=0}^{M} p_k \, h_{k+m} = \sum_{k=0}^{M} p_k \sum_{j=1}^{M} c_j \, z_j^{k+m} = \sum_{j=1}^{M} c_j \, z_j^m \sum_{k=0}^{M} p_k \, z_j^k.$$

Since $p_M = 1$, we rewrite

$$\sum_{k=0}^{M} p_k \, z_j^k = \sum_{k=0}^{M-1} p_k \, z_j^k + z_j^M = p(z_j) = 0,$$

where the last equality follows from eq.(4), since the z_j are the roots of p. Hence

$$\sum_{k=0}^{M} p_k \, h_{k+m} = 0, \qquad \forall \, m \in \{0, \dots, M-1\}.$$

(1.b)

Let $m \in \{0, \dots, M-1\}$. In (5), $p_M = 1$ thus we can isolate the term $k = M$:

$$\sum_{k=0}^{M-1} p_k \, h_{k+m} + h_{M+m} = 0 \quad \Longleftrightarrow \quad \sum_{k=0}^{M-1} h_{m+k} \, p_k = -h_{M+m}.$$

These M equations in the M unknowns $p_0, \dots, p_{M-1}$ can be written as the linear system $Hp = b$, where

$$H = (h_{m+k})_{m,k=0}^{M-1} = \begin{pmatrix} h_0 & h_1 & \cdots & h_{M-1} \\ h_1 & h_2 & \cdots & h_M \\ \vdots & \vdots & \ddots & \vdots \\ h_{M-1} & h_M & \cdots & h_{2M-2} \end{pmatrix} \in \mathbb{C}^{M \times M},$$

$$p = \begin{pmatrix} p_0 \\ p_1 \\ \vdots \\ p_{M-1} \end{pmatrix} \in \mathbb{C}^M, \qquad b = - \begin{pmatrix} h_M \\ h_{M+1} \\ \vdots \\ h_{2M-1} \end{pmatrix} \in \mathbb{C}^M.$$

In other words, $H_{mk} = h_{m+k}$ and $b_m = -h_{M+m}$ for all $m, k \in \{0, \dots, M-1\}$. The system uses all of the $2M$ measurements $h_0, \dots, h_{2M-1}$.

(1.c)

In this question, the rows and columns of all matrices in $\mathbb{C}^{M\times M}$ are indexed by $\{1,\dots,M\}$, as in eq. (7):

$$V_{ik} = z_k^{i-1}, \qquad i,k \in \{1,\dots,M\}.$$

Elements of H with this indexing. By (1.b), for $m,n \in \{0,\dots,M-1\}$, the element of H in row $m+1$ and column $n+1$ equals h_{m+n}. For $i,j \in \{1,\dots,M\}$, we set $m = i-1$ and $n = j-1$, which belong to $\{0,\dots,M-1\}$, and obtain

$$H_{ij} = h_{(i-1)+(j-1)} = h_{i+j-2}, \qquad i,j \in \{1,\dots,M\}. \tag{$*$}$$

Elements of $V\,\mathrm{diag}(c)\,V^T$. Let $i,j \in \{1,\dots,M\}$. By definition of the matrix product,

$$\big(V\,\mathrm{diag}(c)\,V^T\big)_{ij} = \sum_{k=1}^{M}\sum_{l=1}^{M} V_{ik}\,\big(\mathrm{diag}(c)\big)_{kl}\,(V^T)_{lj}.$$

Since $\big(\mathrm{diag}(c)\big)_{kl} = c_k$ if $k=l$ and 0 otherwise, only the terms with $l=k$ remain. Using $(V^T)_{kj} = V_{jk}$, we get

$$\big(V\,\mathrm{diag}(c)\,V^T\big)_{ij} = \sum_{k=1}^{M} V_{ik}\,c_k\,V_{jk} = \sum_{k=1}^{M} c_k\,z_k^{i-1}z_k^{j-1} = \sum_{k=1}^{M} c_k\,z_k^{i+j-2}.$$

Identification. By eq. (3), $z_k^{i+j-2} = \big(\mathrm{e}^{\omega_k}\big)^{i+j-2} = \mathrm{e}^{\omega_k(i+j-2)}$, since the exponent $i+j-2$ is a non-negative integer. Moreover, $i+j-2 \in \{0,\dots,2M-2\}$, so that eq. (2) applies:

$$\big(V\,\mathrm{diag}(c)\,V^T\big)_{ij} = \sum_{k=1}^{M} c_k\,\mathrm{e}^{\omega_k(i+j-2)} \overset{(2)}{=} h_{i+j-2} \overset{(*)}{=} H_{ij}.$$

Since this equality holds for all $i,j \in \{1,\dots,M\}$, we conclude that $H = V\,\mathrm{diag}(c)\,V^T$.

(1.d)

Determinant of H. First, recall that for all $A,B \in \mathbb{C}^{M\times M}$,

$$\det(AB) = \det(A)\det(B) \qquad \text{and} \qquad \det(A^T) = \det(A).$$

Using these two properties and eq. (8):

$$\det H = \det(V)\,\det\big(\mathrm{diag}(c)\big)\,\det(V^T) = \det\big(\mathrm{diag}(c)\big)\,\det(V)^2.$$

For a diagonal matrix, the determinant is the product of the diagonal entries:

$$\det\big(\mathrm{diag}(c)\big) = \prod_{k=1}^{M} c_k.$$

From the hint, V is a Vandermonde matrix, whose determinant is given by

$$\det(V) = \prod_{1\le k<l\le M} (z_l - z_k).$$

Combining it all:

$$\boxed{\det H = \Big(\prod_{k=1}^{M} c_k\Big) \prod_{1\le k<l\le M} (z_l - z_k)^2}$$

which depends only on the $\{c_k\}_{k=1}^M$ and the $\{z_k\}_{k=1}^M$.

Non-singularity of H. The matrix H is invertible if and only if $\det H \neq 0$, i.e. if and only if each of the two factors above is non-zero.

First factor. From the problem statement, $c_k \neq 0$ for all $k \in \{1, \dots, M\}$. Since a product of non-zero complex numbers is non-zero, $\prod_{k=1}^{M} c_k \neq 0$.

Second factor. The factor $\det(V)^2 = \prod_{1 \leq k < l \leq M} (z_l - z_k)^2$ is non-zero if and only if the z_k are pairwise distinct. The problem statement only gives distinct ω_j,and two different exponents can give the same exponential: for instance, $\mathrm{e}^{\omega + 2\pi\mathrm{i}} = \mathrm{e}^{\omega}$. But from the problem statement $\operatorname{Im} \omega_k \in [-\pi, \pi)$, which allows us to conclude.

Let $k, l \in \{1, \dots, M\}$ s.t. $z_k = z_l$, which gives, from eq. (3), $\mathrm{e}^{\omega_k} = \mathrm{e}^{\omega_l}$.

1. $\mathrm{e}^a = \mathrm{e}^b$ if and only if $a - b \in 2\pi\mathrm{i}\,\mathbb{Z}$, so there exists $n \in \mathbb{Z}$ s.t.
$$\omega_k - \omega_l = 2\pi\mathrm{i}\, n.$$
2. Taking the imaginary part of both sides:
$$\operatorname{Im} \omega_k - \operatorname{Im} \omega_l = 2\pi n.$$
3. From the problem statement, $\operatorname{Im} \omega_k, \operatorname{Im} \omega_l \in [-\pi, \pi)$, so
$$-2\pi < \operatorname{Im} \omega_k - \operatorname{Im} \omega_l < 2\pi.$$
4. Hence $-2\pi < 2\pi n < 2\pi$, i.e. $-1 < n < 1$. Since $n \in \mathbb{Z}$, we have $n = 0$.
5. Thus $\omega_k - \omega_l = 0$, i.e. $\omega_k = \omega_l$.
6. Since the ω_j are pairwise distinct by assumption, $k = l$.

Therefore $z_k = z_l$ implies $k = l$: the z_k are pairwise distinct, and $z_l - z_k \neq 0$ for all $1 \leq k < l \leq M$. The second factor is non-zero.

Since both factors are non-zero, $\det H \neq 0$. Hence H is invertible, and the system (6) has a unique solution $p = H^{-1} b$.

(1.e)

Reconstruction of the z_j: From eq.(4) we have

$$p(z) = z^M + \sum_{k=0}^{M-1} p_k\, z^k, \qquad z \in \mathbb{C}.$$

By the fundamental theorem of algebra, p has at least one root $z_1 \in \mathbb{C}$. We can rewrite $p(z) = (z - z_1)\, q_1(z)$ where q_1 is of degree $M - 1$ and whose dominant coefficient equals 1. The degree decreases by one after each iteration and after M iterations, the quotient is of degree 0 and so is equal to 1. We obtain the M roots. From eq.(4), $p(z) = \prod_{j=1}^{M} (z - z_j)$ whose roots are $z_1, \dots, z_M$. Numerically, we compute the roots by finding the eigenvalues of the companion matrix.

Reconstruction of the ω_j: From eq.(3) $z_j = \mathrm{e}^{\omega_j} \neq 0$, we set

$$\omega_j = \operatorname{Log} z_j = \ln |z_j| + \mathrm{i} \arg z_j, \qquad \arg z_j \in [-\pi, \pi),$$

such that $\mathrm{e}^{\omega_j} = z_j$ and $\operatorname{Im} \omega_j \in [-\pi, \pi)$.

Reconstruction of the c_j: Once the z_j are known, the measurements eq.(2) are linear in $c_1, \dots, c_M$. The z_j now only act as known coefficients. Since there are M unknowns, we keep the M first measurements, $k \in \{0, \dots, M-1\}$:

$$h_k = \sum_{j=1}^{M} c_j \, z_j^k, \qquad k \in \{0, \dots, M-1\}.$$

For $k \in \{0, \dots, M-1\}$, eq.(7) gives $V_{k+1,j} = z_j^k$, hence

$$(Vc)_{k+1} = \sum_{j=1}^{M} V_{k+1,j} \, c_j = \sum_{j=1}^{M} z_j^k \, c_j = h_k.$$

These M equalities read

$$\underbrace{\begin{pmatrix} 1 & 1 & \cdots & 1 \\ z_1 & z_2 & \cdots & z_M \\ \vdots & \vdots & \ddots & \vdots \\ z_1^{M-1} & z_2^{M-1} & \cdots & z_M^{M-1} \end{pmatrix}}_{V} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_M \end{pmatrix} = \begin{pmatrix} h_0 \\ h_1 \\ \vdots \\ h_{M-1} \end{pmatrix}.$$

This system admits a solution, namely the vector c that generated the measurements. We now prove its uniqueness.

Uniqueness:

1. *The z_j:* The monic polynomial p of degree M has exactly M roots in $\mathbb{C}$, counted with multiplicity. By eq. (4), these roots are the z_j, and they are simple since the z_j are pairwise distinct, as proved in (1.d). Thus the set $\{z_1, \dots, z_M\}$ is uniquely determined by p.

2. *The ω_j:* Let ω, ω' s.t. $\mathrm{e}^{\omega} = \mathrm{e}^{\omega'} = z_j$ and $\operatorname{Im}\omega, \operatorname{Im}\omega' \in [-\pi, \pi)$. Then $\omega - \omega' = 2\pi\mathrm{i}\, n$ for some $n \in \mathbb{Z}$, and $-2\pi < \operatorname{Im}(\omega - \omega') < 2\pi$, hence $-1 < n < 1$, so $n = 0$ and $\omega = \omega'$.

3. *The c_j:* By (1.d), $\det V = \prod_{1 \le k < l \le M} (z_l - z_k) \neq 0$, so V is invertible. If c and c' are two solutions, then $V(c - c') = 0$, and multiplying on the left by V^{-1} gives $c = c'$. The unique solution is $c = V^{-1}(h_0, \dots, h_{M-1})^T$.

Only the numbering of the z_j is arbitrary: permuting it permutes the columns of V, and the components of c, in the same way. The set of pairs $\{(\omega_j, c_j)\}_{j=1}^{M}$ is therefore uniquely determined.

(1.f)

Implementation: The method is implemented in the function `prony` (file `prony.py`), which takes as input the $2M$ measurements $h_0, \dots, h_{2M-1}$ and returns the ω_j and c_j. It follows the four steps of (1.b)–(1.e):

1. Build H and b from the measurements, then solve $Hp = b$ (possible and unique by (1.d)).

2. Compute the roots $z_1, \dots, z_M$ of the polynomial $z^M + \sum_{k=0}^{M-1} p_k z^k$, as the eigenvalues of its companion matrix.

3. Set $\omega_j = \operatorname{Log} z_j$ for $j \in \{1, \dots, M\}$.

4. Solve the system $Vc = (h_0, \dots, h_{M-1})^T$.

Validation: We test the implementation on signals whose parameters are known, for $M \in \{2, 5, 10\}$. Prony's method returns the pairs (ω_j, c_j) in an arbitrary order, so we sort both the true and the recovered pairs by increasing $\operatorname{Im} \omega_j$ before comparing them. The results are given in Table 1.

M	$\max_j \lvert\omega_j - \hat{\omega}_j\rvert$	$\max_j \lvert c_j - \hat{c}_j\rvert$
2	1.3e-14	6.1e-14
5	1.3e-14	1.7e-14
10	2.3e-10	5.0e-09

Table 1: (1.f) Recovery errors of Prony's method on synthetic signals.

The parameters are recovered up to machine precision. For $M = 10$, the error is larger because H becomes ill-conditioned: it contains the Vandermonde matrix V, whose conditioning quickly worsens as M grows (course notes, §B.2.4).

2 Non-uniform sampling

(2.a)

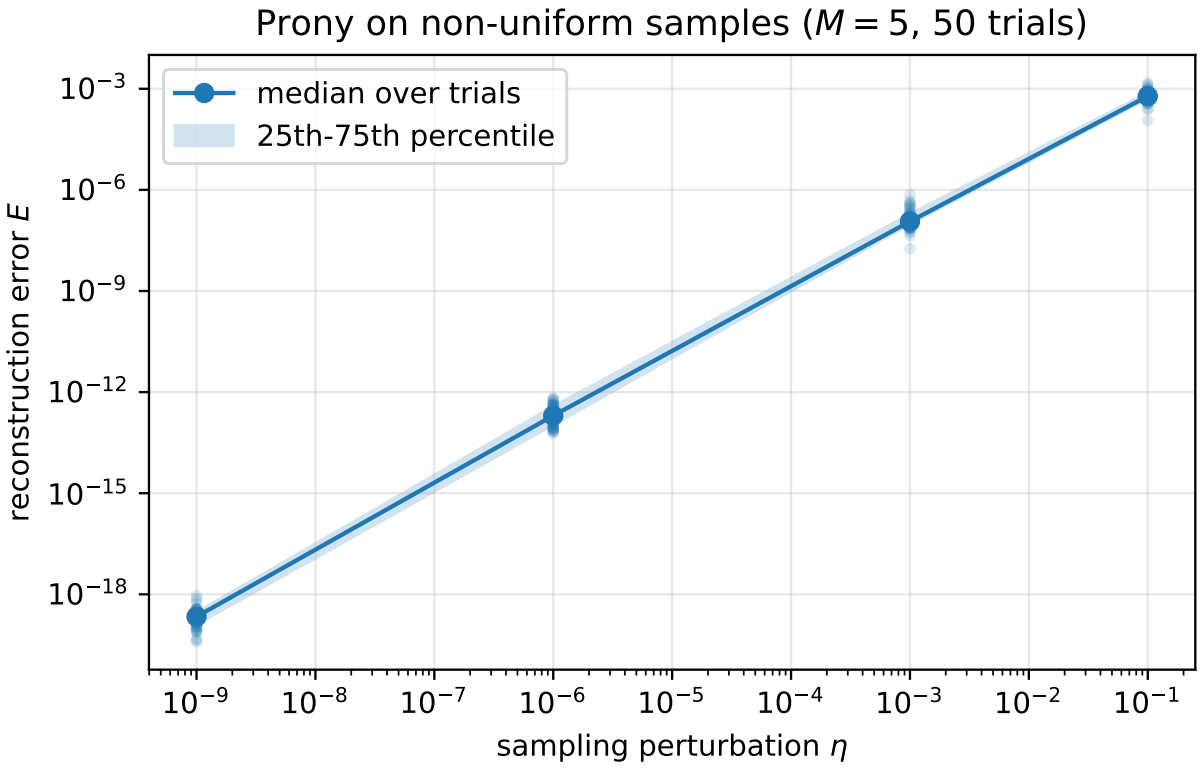


Figure 1: Reconstruction error $\mathcal{E}$ versus η (median over 50 draws).

Observations: The parameters ω_j and c_j are drawn once with a fixed seed. The sampling error is drawn 50 times and the median of $\mathcal{E}$ is reported, for a more robust analysis. The band is narrow, hence the behaviour does not depend much on a particular draw. We can observe from the figure that the error $\mathcal{E}$ is proportional to η^2 .The slope of the straight line is ≈ 1.93.

Origin of the error: Denote by $\tilde{h}_k := h(k+\varepsilon_k)$ the perturbed measurements. With $z_j = e^{\omega_j}$, eq. (2) becomes

$$\tilde{h}_k = \sum_{j=1}^{M} (c_j\, z_j^{\varepsilon_k})\, z_j^k, \qquad k \in \{0, \dots, 2M-1\}.$$

The coefficients become $c_j z_j^{\varepsilon_k}$ and therefore depend on k: each measurement has its own set of coefficients. Repeating the computation of (1.a), for $m \in \{0, \dots, M-1\}$,

$$\sum_{k=0}^{M} p_k\, \tilde{h}_{k+m} = \sum_{j=1}^{M} c_j\, z_j^m \sum_{k=0}^{M} p_k\, z_j^k\, z_j^{\varepsilon_{k+m}}.$$

Because the factor $z_j^{\varepsilon_{k+m}}$ depends on k, it cannot be taken out of the inner sum, which is therefore no longer equal to $p(z_j) = 0$. Relation (5) is no longer satisfied:

$$\sum_{k=0}^{M} p_k\, \tilde{h}_{k+m} \neq 0 \quad \text{in general.}$$

This directly affects the computation of the vector p in eq. (6). From the perturbed measurements, we obtain a different matrix H' and a different vector b'. Hence $p' = H'^{-1} b' \neq p$, and the polynomial $z^M + \sum_k p'_k z^k$ no longer has the z_j as roots. The new roots z'_j give new exponents ω'_j and a new Vandermonde matrix, which in turn gives new coefficients c'_j.

The inaccuracy thus enters at the first step of the method, the computation of p from (6).

(2.b)

Preprocessing with an interpolation: Previously, the measurements, $\tilde{h}_k = h(k+\varepsilon_k)$, were used directly and so relation (5) was unsatisfied. From the data $\{(k+\varepsilon_k, \tilde{h}_k)\}_{k=0}^{2M-1}$, we build an interpolant $\hat{h}$ and evaluate it at the integer $k = 0, ..., 2M-1$. These values are then used in Prony's method. Hence the error on the roots is no longer caused by the sampling shift anymore but by the quality of the interpolant.

Formally for $\delta_k := \hat{h}(k) - h(k)$ we have

$$\sum_{k=0}^{M} p_k\, \hat{h}(k+m) = \sum_{k=0}^{M} p_k\, h(k+m) + \sum_{k=0}^{M} p_k\, \delta_{k+m} = \sum_{k=0}^{M} p_k\, \delta_{k+m}.$$

If $\delta_k = 0$ for all k, eq.(5) holds exactly and the recovery is exact.

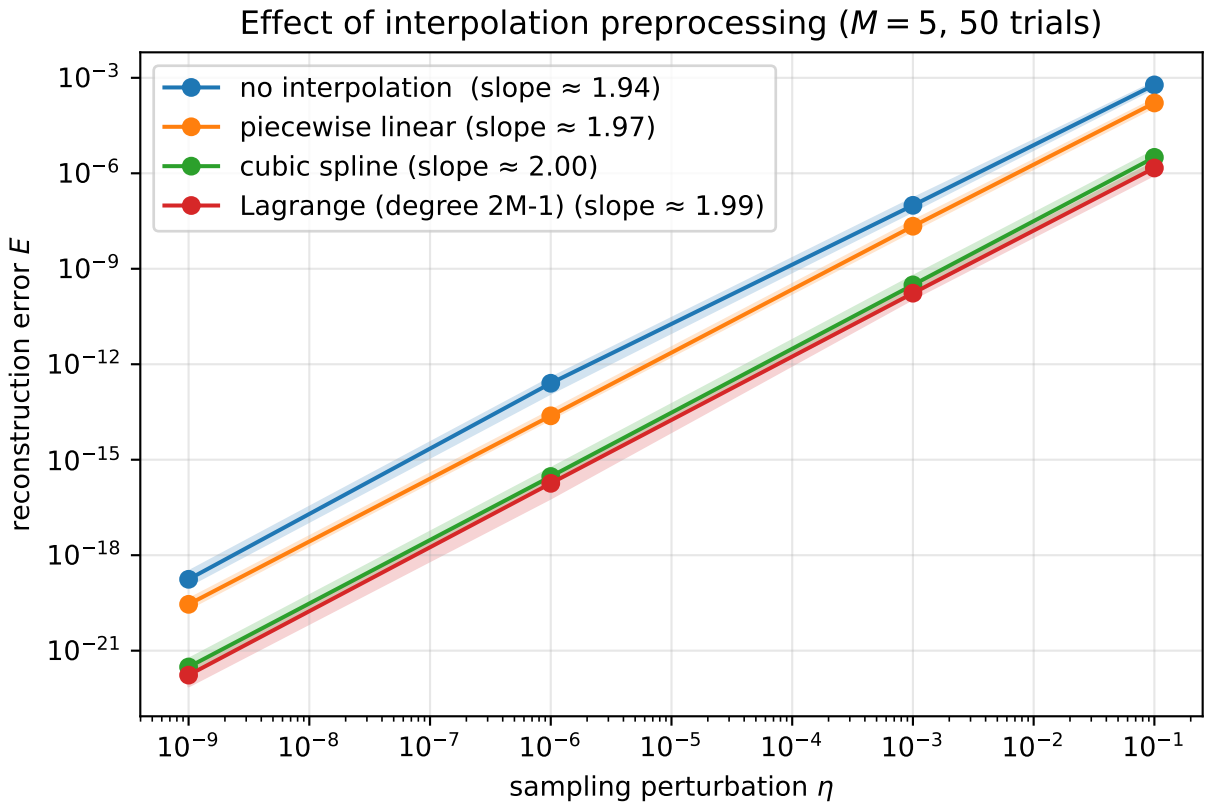


Figure 2: Reconstruction error $\mathcal{E}$ versus η, with and without interpolation preprocessing (median over 50 draws).

Choice of the interpolant: Once again, the same procedure (50 different draws of ε) is applied for a more robust analysis. The bands remain narrow for every method, so the conclusions do not depend on a particular draw.

1. *Piecewise linear interpolation* (`numpy.interp`). On each interval $[x_{i-1}, x_i]$, $\hat{h}$ is the straight line through the two neighbouring data points,

$$\hat{h}(x) = \frac{x_i - x}{x_i - x_{i-1}}\, y_{i-1} + \frac{x - x_{i-1}}{x_i - x_{i-1}}\, y_i.$$

   By (B.1) with $n = 1$, its error on an interval of length $\Delta = x_i - x_{i-1}$ is bounded by $\frac{\Delta^2}{8} \max |h''|$, here $\Delta = 1$. Since the data points are one unit apart, straight lines between them cannot follow the curvature of h. Moreover, at the two ends of the interval, `numpy.interp` does not extend the line but simply repeats the boundary value. As a result, the error is only reduced by a factor of 3 to 8, which is less than one order of magnitude.

2. *Cubic spline* (`scipy.interpolate.CubicSpline`): For smooth functions, the error of cubic spline interpolation behaves as $O(\Delta^4)$ (see course notes C.8.1). Compared with the direct approach, the error is reduced by a factor of nearly 10^3 for $\eta = 10^{-9}$, and this gap narrows as η increases.

3. `Global Lagrange polynomial of degree` $2M - 1$: A single polynomial passes through all $2M$ data points. It is evaluated with the barycentric formula . Its error is controlled by (B.1) with $n = 2M - 1$. With equispaced nodes, high-degree polynomial interpolation can suffer from the Runge phenomenon. Here there are only 10 nodes, so it is not a concern and the derivatives of h do not grow too quickly(sum of exponentials). This method has the smallest error, slightly better than the cubic spline .

Interpolation does not allow an exact recovery of the parameters. The error still scales as η^2, but its magnitude is reduced by up to three orders of magnitude with a high-order interpolant.

3 Recovery of a piecewise constant function

(3.a)

We want to write eq. (10),

$$\mathcal{F}\{f\}(\gamma) := \hat{f}(\gamma) = \sum_{j=1}^{M} \frac{b_j}{i\gamma} \left(e^{-i\gamma T_j} - e^{-i\gamma T_{j+1}} \right),$$

in the form of eq. (2),

$$h_k := h(k) = \sum_{j=1}^{M} c_j \exp(\omega_j k), \qquad k = 0, \ldots, 2M-1.$$

Splitting the sum, each T_j with $2 \le j \le M$ appears twice, with coefficients b_j and $-b_{j-1}$, while T_1 and T_{M+1} appear only once (this is easily seen by writing the terms for $M = 2$). Regrouping the terms with the same exponential, we get

$$\hat{f}(\gamma) = \frac{1}{i\gamma} \sum_{j=1}^{M+1} (b_j - b_{j-1})\, e^{-i\gamma T_j}, \qquad \text{with } b_0 = b_{M+1} = 0.$$

Multiplying by $i\gamma$, we denote

$$\hat{h}(\gamma) := (i\gamma)\hat{f}(\gamma) = \sum_{j=1}^{M+1} (b_j - b_{j-1})\, e^{-i\gamma T_j}.$$

Applying this to the K measurements $\{\hat{f}(k\Delta\gamma)\}_{k=0}^{K-1}$, we get, for $k = 0, \ldots, K-1$,

$$\hat{h}_k := \hat{h}(k\Delta\gamma) = (ik\Delta\gamma)\hat{f}(k\Delta\gamma) = \sum_{j=1}^{M+1} (b_j - b_{j-1})\, e^{-ik\Delta\gamma\, T_j}.$$

This is exactly the form of eq. (2),

$$\hat{h}_k = \sum_{j=1}^{M'} c_j\, e^{\omega_j k},$$

with $M' = M + 1$ exponentials and

$$c_j = b_j - b_{j-1}, \qquad \omega_j = -i\,\Delta\gamma\, T_j, \qquad j \in \{1, \ldots, M+1\}.$$

The problem statement requires $c_j \neq 0$. Hence, we must have $b_j \neq b_{j-1}$ for $j \in \{1, \ldots, M+1\}$ (in particular $b_1 \neq 0$ and $b_M \neq 0$).

The unknowns are therefore the $M+1$ breakpoints T_j and the M values b_j. Once Prony's method has recovered the ω_j and c_j, we obtain

$$T_j = \frac{i\,\omega_j}{\Delta\gamma} = -\frac{\operatorname{Im}\omega_j}{\Delta\gamma}, \qquad b_j = \sum_{l=1}^{j} c_l,$$

the second relation being a telescoping sum (with the pairs sorted by increasing T_j). The equivalence holds under the assumptions of Part 1, namely $c_j \neq 0$ and $\omega_j \in [-\alpha, 0] + i[-\pi, \pi)$.

(3.b)

Minimal number of measurements: By (3.a) we have that $\hat{h}_k$ is a sum of $M' = M + 1$ exponentials. Prony's method with M' exponentials requires the system (6), in which H is of size $M' \times M'$ with entries $H_{mk} = \hat{h}_{m+k}$, $m, k \in \{0, \dots, M' - 1\}$, and $b_m = -\hat{h}_{M'+m}$. The first row of H uses $\hat{h}_0, \dots, \hat{h}_{M'-1}$, each following row adds one new measurement, up to $\hat{h}_{2M'-2}$, and b adds $\hat{h}_{2M'-1}$. Hence

$$K = 2M' = 2M + 2.$$

This is minimal, since there are $2M' = 2M + 2$ unknowns (the ω_j and the c_j).

Condition on $\Delta\gamma$. In the problem statement, we had the following hypothesis $\omega_j \in [-\alpha, 0] + i[-\pi, \pi)$. Here $\operatorname{Re}\omega_j = 0$, and with $\Delta\gamma > 0$ and $T_j \geq 0$,

$$-\pi \leq \operatorname{Im}\omega_j < \pi \iff -\pi \leq -\Delta\gamma\, T_j < \pi \iff \Delta\gamma\, T_j \leq \pi,$$

the right inequality being always satisfied since $-\Delta\gamma T_j \leq 0$. This must hold for all $j \in \{1, \dots, M+1\}$ and since $T_1 < \cdots < T_{M+1}$, the most restrictive case is $j = M + 1$, so that

$$0 < \Delta\gamma \leq \frac{\pi}{T_{M+1}}.$$

This ensures that two different points T_j never give the same root $z_j = e^{\omega_j}$, whose phase is only defined modulo 2π (see 1.d).

Under these conditions (and $b_j \neq b_{j-1}$, see 3.a), H is invertible (1.d), the ω_j and c_j are recovered exactly (1.e), and the T_j and b_j follow from (3.a).

(3.c)

j	T_j	$\hat{T}_j$	$\lvert T_j - \hat{T}_j \rvert$	b_j	$\hat{b}_j$	$\lvert b_j - \hat{b}_j \rvert$
1	0.5	0.500000	1.4e-14	1.0	1.000000	1.8e-13
2	1.3	1.300000	2.4e-13	-0.5	-0.500000	3.6e-14
3	2.0	2.000000	1.8e-13	2.0	2.000000	5.0e-14
4	3.2	3.200000	1.5e-13	0.8	0.800000	7.7e-14
5	4.0	4.000000	6.2e-15	–	–	–

Table 2: (3.c) True and recovered parameters ($M = 4$, $K = 10$, $\Delta\gamma = 0.8\pi/T_5$).

Comments:

The errors are of the order of machine precision: we correctly recover $\{b_j\}_{j=1}^4$ and $\{T_j\}_{j=1}^5$

(3.d)

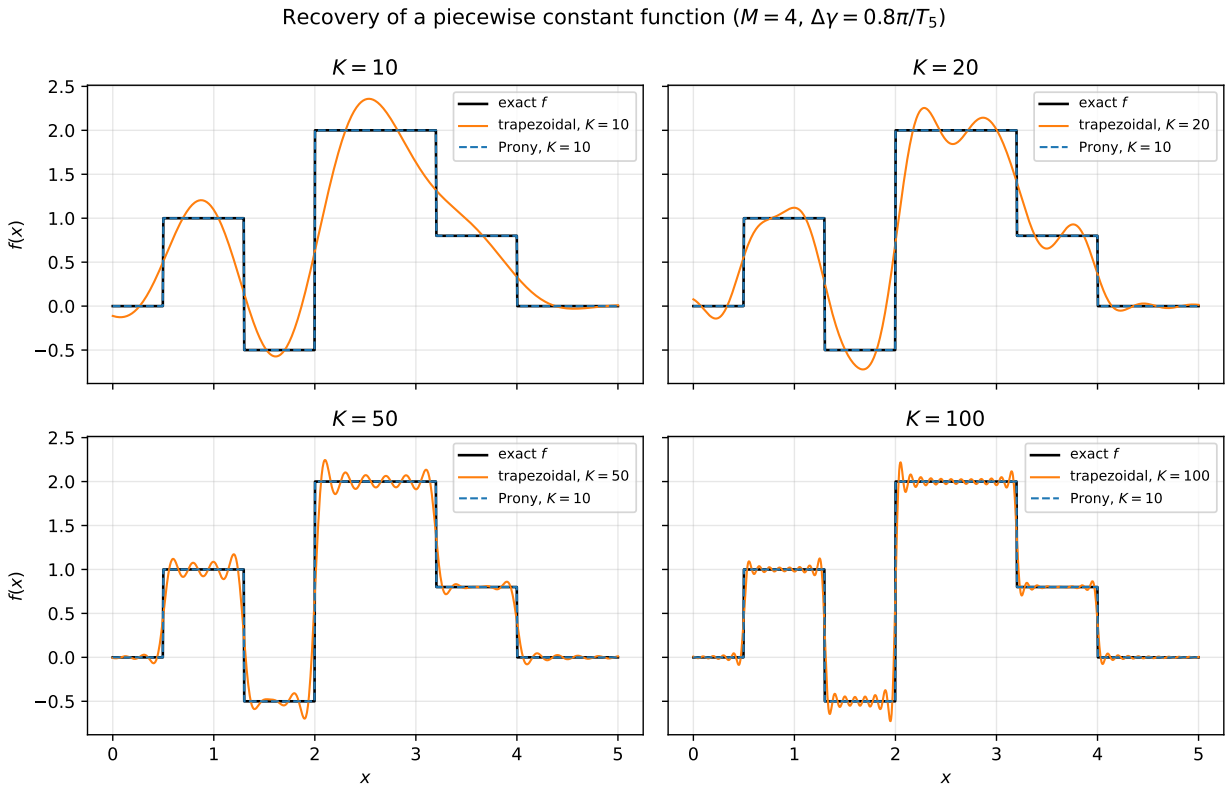


Figure 3: Exact function f, trapezoidal reconstructions for $K \in \{10, 20, 50, 100\}$, and Prony's reconstruction with $K = 10$ measurements.

We keep the same T_j, b_j and $\Delta\gamma = 0.8\pi/T_5$ as in (3.c). Hence, increasing K means using higher frequencies, up to $(K-1)\Delta\gamma$. The value $\hat{f}(0)$ is computed separately as the area under f, $\hat{f}(0) = \sum_j b_j(T_{j+1} - T_j)$.

Observations: From Figure 3, Prony's reconstruction matches f exactly with only 10 measurements (mean error $\approx 6 \cdot 10^{-14}$). The trapezoidal reconstruction is much worse. For $K = 10$, it gives a smooth curve that barely follows the different plateaus. As K increases, the shape gets better and the mean error decreases, from 0.24 for $K = 10$ to 0.04 for $K = 100$. But oscillations appear around each jump : they get narrower but never disappear, and the maximum error stays above 1 even for $K = 100$.

Why Prony succeeds: Thanks to Prony's method and to the fact that the form of $\hat{h}$ is known (a sum of $M+1$ exponentials, see (3.a)), we only have to recover $2M+2$ parameters, the ω_j and the c_j. By (3.b), $2M + 2 = 10$ exact measurements are enough to recover them, and hence to reconstruct f exactly. However, as seen in Part 2, the method is sensitive to errors in the measurements.

Why the trapezoidal rule struggles: The trapezoidal rule replaces the integral

$$f(x) = \frac{1}{\pi} \int_0^\infty \Re\{\hat{f}(\gamma)\, e^{i\gamma x}\}\, d\gamma$$

by a finite sum, and therefore keeps only the frequencies up to $(K-1)\Delta\gamma$. The higher frequencies are lost, although they are needed to reproduce the sharp jumps of f. As $K \to \infty$, the sum covers the whole integral and the reconstruction converges to f (except at the jumps), but this would

require infinitely many measurements, which is impossible. The method is therefore limited by its number of measurements. Moreover, it makes no a priori assumption on f: it does not use the fact that f is piecewise constant. This explains its much larger error.

LLM

An LLM was used for additionnal hints, in exercice (1.d), (1.e) and 3.a.

It was also used to correct my code after a first attempt an to make the different plots.

It helped with the wording of the sentence and the restructuration of some of my arguments. In Part 2, it helped with writing the code and explained the different interpolants.